

Type Checking

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Type Checking Algorithm

- Assume, we have a *simple* expression e .
- In an environment E , there is at most one type t such that $E \vdash e : t$
- A **type checking algorithm** constructs the derivation $E \vdash e : t$ given E and e .
- The algorithm proceeds in a *syntax-directed fashion* i.e., given the simple expression e , the algorithm can identify the typing rule for e and recursively evaluate the premises of the rule.

Type Checking – Simple Expressions

IF e_1 THEN e_2 ELSE e_3

$$\frac{E \vdash e_1 : \text{bool} \quad E \vdash e_2 : t \quad E \vdash e_3 : t}{E \vdash \text{IF } e_1 \text{ THEN } e_2 \text{ ELSE } e_3 : t}$$

1. For the above derivation, the algorithm recurses over the subproblems (E, e_1) , (E, e_2) , (E, e_3) .
2. Successfully evaluates the subproblems with types t_1, t_2, t_3 .
3. Checks $t_1 = \text{bool}$ and $t_2 = t_3$.
4. If yes, we have a derivation for $E \vdash e : t$. If no, throw one of the following errors:
 - e_1 is not of type bool.
 - The types of e_2 & e_3 are different.

- An expression e is **open** if some variables occurs **free** in e .
- Given an expression e , we call a variable x **free in** e if x has an occurrence in e that is not in scope.

Example:

- *Open Expression*: $\text{LET } f = \lambda x. fx \text{ IN } f$
- *Closed Expression*: $\text{LET REC } fx = fx \text{ IN } f$
- In the expression $\text{LET } f = \lambda x. fx \text{ IN } f$, the first occurrences of f and x are **binding occurrences** and all other are **using occurrences**.

Binding Rules (1/3)

- Binding rules are simplifications of typing rules.
- Typing rules for expressions follow binding rules for variables.
- Assume, X denote the finite set of variables. An expression e is closed in X if every variable that is **free** in e is an element of X .

Binding Rules (2/3)

$$\frac{}{X \vdash c} \quad \frac{x \in X}{X \vdash x} \quad \frac{X \vdash e_1 \quad X \vdash e_2}{X \vdash e_1 \text{ o } e_2} \quad \frac{X \vdash e_1 \quad X \vdash e_2}{X \vdash e_1 e_2}$$

$$\frac{X \vdash e_1 \quad X \vdash e_2 \quad X \vdash e_3}{X \vdash \text{IF } e_1 \text{ THEN } e_2 \text{ ELSE } e_3}$$

$$\frac{X \vdash e_1 \quad \dots \quad X \vdash e_n}{X \vdash (e_1, \dots, e_n)} \quad \frac{X \vdash e}{X \vdash \pi_i e}$$

Binding Rules (3/3)

$$\frac{X \vdash e_1 \quad X, x \vdash e_2}{X \vdash \text{LET } x = e_1 \text{ IN } e_2}$$

$$\frac{X, x \vdash e}{X \vdash \lambda x. e}$$

$$\frac{X, f, x \vdash e_1 \quad X, f \vdash e_2}{X \vdash \text{LET REC } f x = e_1 \text{ IN } e_2}$$



This is the End

